

ALEX R. TECHMAN

AI Engineer | Machine Learning Specialist Ottawa, ON | (555) 012-3456 | alex.techman.ai@email.com

[\[LinkedIn.com/in/alex-ai-engineer\]](#) | [\[GitHub.com/alex-tech-ai\]](#)

PROFESSIONAL SUMMARY

Innovative AI Engineer with 8+ years of experience specializing in Agentic AI, Retrieval-Augmented Generation (RAG), and Large Language Model (LLM) orchestration. Proven track record of deploying production-ready AI agents using LangChain and CrewAI. Expert in building scalable MLOps pipelines on AWS and Azure Databricks.

CORE COMPETENCIES

- **AI/ML:** LLMs (GPT-4, Claude 3, Llama 3), RAG Pipelines, Vector Databases (Pinecone, Milvus), Prompt Engineering.
- **Frameworks:** PyTorch, TensorFlow, LangChain, CrewAI, AutoGen, Haystack.
- **Languages & Tools:** Python (Expert), SQL, Bash, Docker, Kubernetes, Git.
- **Cloud & MLOps:** AWS (S3, Glue, SageMaker), Azure Databricks, MLflow, CI/CD.

PROFESSIONAL EXPERIENCE

Lead AI Engineer | NextGen Solutions *Ottawa, ON | Jan 2022 – Present*

- Architected an **Agentic AI workflow** using CrewAI that reduced manual data entry time by 70% for legal clients.
- Implemented an advanced **RAG system** utilizing hybrid search (semantic + keyword) to query 1M+ internal documents with 95% accuracy.
- Optimized model inference costs by 40% through the implementation of quantization and strategic model routing.
- Managed the end-to-end MLOps lifecycle on **Azure Databricks**, ensuring 99.9% uptime for production models.

Machine Learning Engineer | Innovate Corp *Toronto, ON | May 2018 – Dec 2021*

- Developed and deployed a real-time recommendation engine using PyTorch that increased user engagement by 25%.
- Built automated ETL pipelines using **AWS Glue and S3** to process 5TB of daily streaming data.

- Collaborated with cross-functional teams using **Scrum/Agile** methodologies to deliver bi-weekly feature updates.
- Authored technical documentation in Confluence and managed task tracking via Jira.

PROJECTS

Multi-Agent Research Assistant (Open Source) * Developed a GitHub repository using **LangChain** and **MCP (Model Context Protocol)** to create a swarm of agents that research, summarize, and cite academic papers.

- Integrated a Streamlit UI for real-time interaction.

Self-Hosted LLM Environment * Configured a localized development environment using **WSL2 (Ubuntu)** and Docker to run Llama 3 models offline, ensuring data privacy for sensitive projects.

EDUCATION

Master of Science in Computer Science (Specialization in AI) *University of Ottawa, Ottawa, ON*

Bachelor of Engineering in Software Engineering *Carleton University, Ottawa, ON*

CERTIFICATIONS

- AWS Certified Machine Learning – Specialty
- Databricks Certified Professional Data Scientist
- DELF B2 (French Language Proficiency)